

Name: Diksha Dipak Jadhav

Q1. A Food based Startup wants to setup its internal network. Answer the following

- a) Explain the difference between public IP and private IP addresses. Give an example of when each is used

Answer: **Difference between Public IP and Private IP Address**

Category	Public IP Address	Private IP Address
Scope	Has a global reach.	Only local reach.
Communication	Used to communicate over the Internet.	Used to communicate within a private network.
Uniqueness	A unique address not used by other devices on the Internet.	An address from a smaller range used by other devices in other local networks.
Origin	Assigned by the ISP.	Assigned by the router to a specific device on the local network.
Range	Any IP not in the private IP address ranges.	Any address from the following ranges: 10.0.0.0 - 10.255.255.255; 172.16.0.0 - 172.31.255.255; 192.168.0.0 - 192.168.255.255
Security	Public IPs require additional security measures, they are prone to attacks.	Private IPs are secure.
Example	82.129.73.111	10.0.0.0

☐ **Example Use Case for Public IP Address:**

- A startup's website hosted on a server will have a public IP so customers can access it from anywhere in the world.
- Example: 203.0.113.25

☐ **Example Use Case for Private IP Address:**

- Employees' laptops, printers, and internal servers in the startup's office will use private IPs to connect to each other.
- Example ranges:
- 192.168.x.x
- 10.x.x.x
- 172.16.x.x – 172.31.x.x

- b) **What is a CIDR block? If the company has been assigned 192.168.0.0/24, how many usable IP addresses are available?**

Answer:

CIDR Block and Usable IP Addresses

A **CIDR block** (Classless Inter-Domain Routing) is a way of defining a range of IP addresses using a network prefix. It helps in efficient use of IP addresses by specifying how many bits are used for the network part and how many are for host devices.

For the CIDR block **192.168.0.0/24**:

- “/24” means **24 bits are used for the network**
- Remaining **8 bits** are used for hosts
- Total IP addresses = $2^8 = 256$

Out of these:

- 1 IP is reserved for the **network address** (192.168.0.0)
- 1 IP is reserved for the **broadcast address** (192.168.0.255)

So, **usable IP addresses = $256 - 2 = 254$**

Therefore, the company can use 254 IP addresses for its internal devices.

c) **If a company wants to block a few websites so that employees cannot visit them using the company’s internet, where can this configuration be done while setting up the network?**

Answer:

Blocking Websites in a Company Network:

To block certain websites so employees cannot access them through the company’s internet, the configuration is typically done at the **network level**, specifically in one of these places:

➤ **Where to Configure Website Blocking**

- **Firewall**
 - Most enterprise firewalls allow administrators to create rules that block traffic to specific domains or IP addresses.
 - This is the most common and secure method since all traffic passes through the firewall.
- **Router / Gateway**
 - Many business-grade routers have built-in content filtering features.
 - Administrators can configure access control lists (ACLs) or URL filtering directly on the router.
- **Proxy Server**
 - If the company uses a proxy server, it can be configured to deny requests to certain websites.
 - This is useful for monitoring and controlling web traffic more granularly.
- **DNS Filtering**
 - By using DNS services (like OpenDNS or Cloudflare Gateway), requests to blocked domains can be intercepted and denied.
 - This is a lightweight option that doesn’t require heavy firewall rules.

Q2. Draw a simple diagram showing:

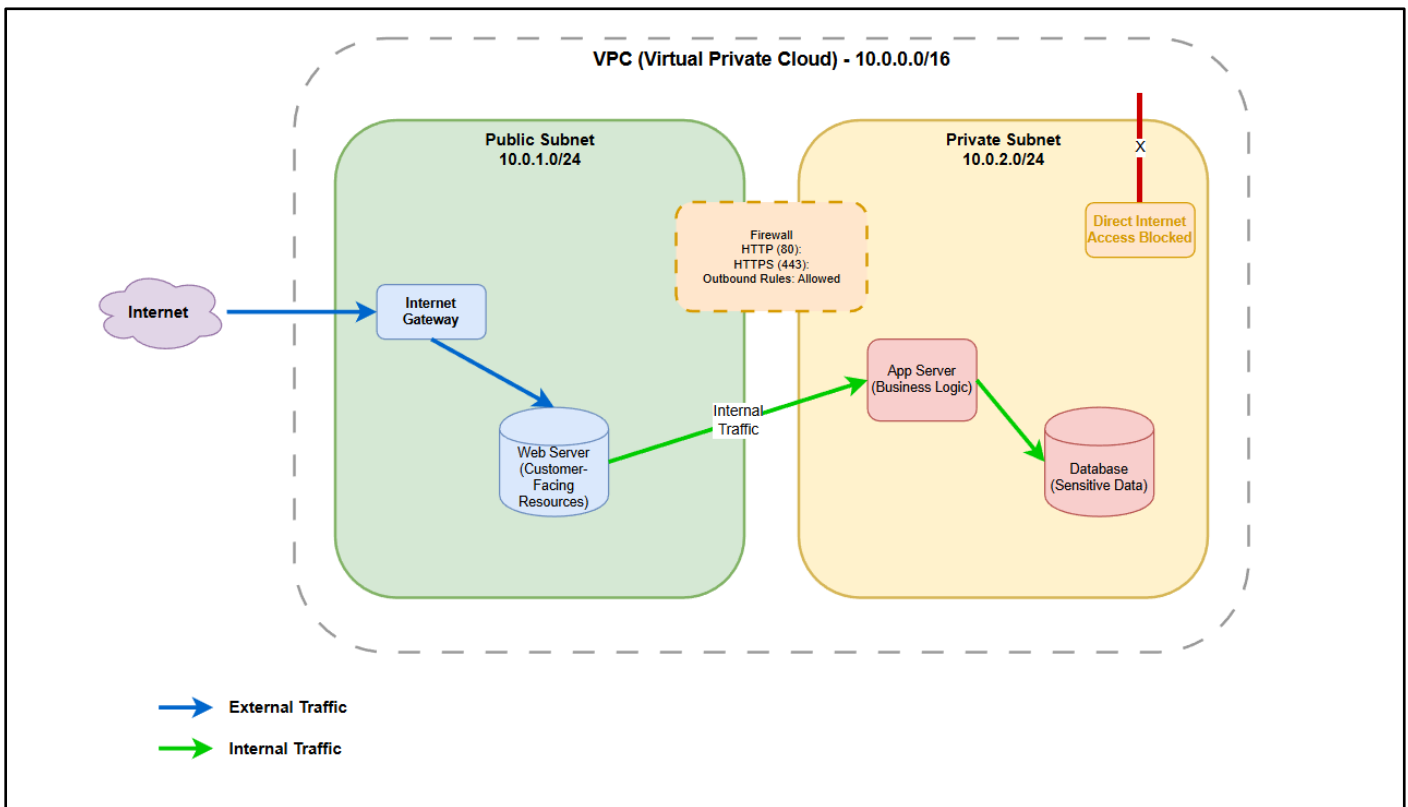
A **VPC (Virtual Private Cloud)** is a private network created inside the cloud where we can launch our resources securely.

In this setup, the VPC has the IP range **10.0.0.0/16**.

This IP range is divided into two subnets:

Public Subnet: 10.0.1.0/24

Private Subnet: 10.0.2.0/24



a. Which data of an organization will reside in public and private subnet.

Ans:1. Public Subnet

The public subnet is used for internet-facing resources.

Resources in public subnet:

- 1) Internet Gateway
- 2) Web Server or Bastion Host

Type of data stored:

- Website content
- Static files
- Public APIs
- Non-sensitive information

This subnet allows incoming traffic from the internet.

2. Private Subnet

The private subnet is used for internal and sensitive resources.

Resources in private subnet:

Application Server
Database Server

Type of data stored:

- Customer data
- User details

- Business and financial data
- Internal application logic

This subnet does **not allow direct internet access**.

b. Firewall with inbound and outbound rules to configure access

Ans: Firewall Configuration

Firewall rules (Security Groups / NACLs) are used to control access.

Inbound Rules:

- 1) Allow HTTP (Port 80)
- 2) Allow HTTPS (Port 443)

Outbound Rules:

- 1) All outbound traffic is allowed
- 2) Firewall ensures that only authorized traffic can enter or leave the network.

c. How User Reaches the Private Subnet:

Ans: A private subnet cannot be accessed directly from the internet for security reasons. Users reach it indirectly through a public component.

1. **User (Internet)**
 - User sends a request from a browser or application.
2. **Internet Gateway (IGW)**
 - The request first enters the **VPC** through the Internet Gateway.
3. **Public Subnet**
 - The request reaches a resource in the **public subnet** (e.g., Load Balancer, Bastion Host, or Web Server).
4. **Firewall / Security Groups**
 - Firewall rules allow only specific traffic (like HTTP/HTTPS or SSH).
5. **Private Subnet**
 - The public subnet resource forwards the request to resources in the **private subnet** (e.g., Application Server, Database).
6. **Response Back**
 - The response travels back through the same path to the user.